



(a) Semelparity: one-time reproduction. An agave (*Agave americana*) is an example of semelparity. The leaves of the plant are visible at the base of the giant flowering stalk, which is produced only at the end of the agave's life.

▼ **Figure 53.12 Semelparity and iteroparity.**



(b) Iteroparity: repeat reproduction. Organisms that reproduce repeatedly, such as the bur oak (*Quercus macrocarpa*), undergo iteroparity. One oak tree can produce thousands of acorns (inset) per year over the course of many decades.

In contrast to semelparity is **iteroparity** (from the Latin *iterare*, to repeat), or repeated reproduction. As an example, a female loggerhead turtle produces four clutches totaling approximately 300 eggs in a year. It then typically waits two to three years before laying more eggs; presumably, the turtles lack sufficient resources to produce that many eggs every year. A mature turtle may lay eggs for 30 years after the first clutch. Horses and other large mammals also reproduce repeatedly, as do many fish, sea urchins, and long-lived trees, such as maples and oaks (Figure 53.12b).

Finally, organisms also vary in how many offspring they produce. Some species, such as the white rhinoceros (see Figure 53.11), produce a single calf when they reproduce, while most insects and many plants produce large numbers of offspring. Such variation in offspring number has other consequences as well; as you'll read, a species that produces one or a few offspring may provision them better than does a species that produces many offspring.

"Trade-offs" and Life Histories

No organism could produce thousands of offspring yet provision each of them as well as does a white rhinoceros caring for its single calf. There is a trade-off between the number of offspring and the amount of resources a parent can devote to each offspring. Such trade-offs occur because organisms do not have access to unlimited amounts of resources. As a result, the use of resources for one function (such as reproduction) can reduce

the resources available for supporting another function (such as survival). In Eurasian kestrels, for example, caring for a larger number of young lowered the survival rates of the parents (Figure 53.13). In another study, in Scotland, researchers found that female red deer that reproduced in a given summer were more likely to die the next winter than were females that did not reproduce.

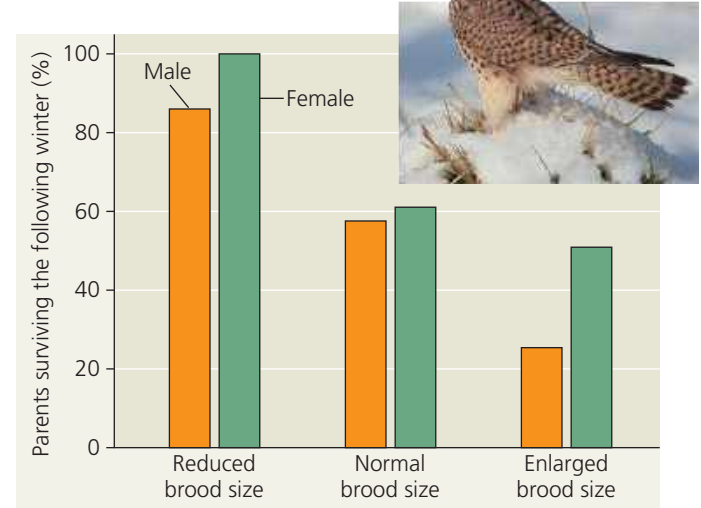
Selective pressures also influence trade-offs between the number and size of offspring. Plants and animals whose young have a low chance of survival often produce many small offspring. For example, plants that colonize disturbed environments usually produce many small seeds, only a few of which may reach a suitable habitat. Small size may also increase the chance of seedling establishment by enabling the seeds to be carried longer

▼ **Figure 53.13 Inquiry**

How does caring for offspring affect parental survival in kestrels?

Experiment Cor Dijkstra and colleagues in the Netherlands studied the effects of parental caregiving in Eurasian kestrels over five years. The researchers transferred chicks among nests to produce reduced broods (three or four chicks), normal broods (five or six), and enlarged broods (seven or eight). They then measured the percentage of male and female parent birds that survived the following winter. (Both males and females provide care for chicks.)

Results



Conclusion The lower survival rates of kestrels with larger broods indicate that caring for more offspring negatively affects survival of the parents.

Data from C. Dijkstra et al., Brood size manipulations in the kestrel (*Falco tinnunculus*): effects on offspring and parent survival, *Journal of Animal Ecology* 59:269–285 (1990).

INTERPRET THE DATA The males of some bird species provide no parental care. If this were true for the Eurasian kestrel, predict how the experimental results would differ from those shown here.